Python Automation, Scheduling, Logging, and Web Scraping

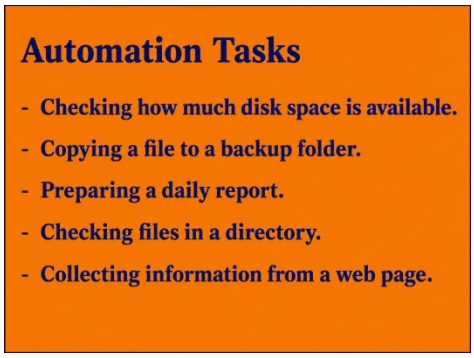
What is Automation?

Automation means writing instructions that perform a task with less manual effort. In DevOps, repeated tasks such as checking disk space, backing up files, preparing daily reports and collecting data from web pages are automated to save effort and avoid mistakes. **Python** performs the steps, a **scheduler** decides when the script runs, and **logs** record what happened.

Before automating a task, define:

- **Inputs:** what information the script receives.
- **Sources and Outputs:** which files or folders it uses, and where results are saved.
- **Error Handling:** what happens if something is missing.
- **Idempotency:** what happens when the script runs repeatedly.

Workflow: Read Input → Check Input → Perform Task → Save Result → Record Log



Practice Setup and Paths (pathlib)

Work in a separate folder with an isolated virtual environment (Python 3 on Ubuntu). The pathlib module handles file paths as objects, and the / operator joins path parts safely.

```
mkdir week4 && cd week4
python3 -m venv .venv
source .venv/bin/activate
```

Method	Purpose
exists()	Checks whether the path exists.
is_file() / is_dir()	Checks for a regular file / a directory.
mkdir()	Creates a directory (parents=True, exist_ok=True).
iterdir() / glob()	Lists directory contents / matches a pattern such as *.txt.
stat()	Returns file metadata such as st_size.

Create a separate folder and set up an isolated virtual environment:

1. Create a new folder and navigate into it:

```
$ mkdir week4
$ cd week4
```

2. Create and activate virtual environment:

```
$ python3 -m venv .venv
$ source .venv/bin/activate
```

Example: Listing .txt Files with pathlib

This short script loops over every .txt file in the input folder and prints each name with its size in bytes.

```
from pathlib import Path

folder = Path("input")

for file_path in sorted(folder.glob("*.txt")):
    if file_path.is_file():
        size = file_path.stat().st_size
        print(f"{file_path.name}: {size} bytes")
```

How it works

- Path("input") creates a path object for the folder.
- sorted(folder.glob("*.txt")) finds matching files in a predictable order.
- is_file() skips folders; stat().st_size returns the size in bytes.

```
Sample output:
alpha.txt: 5 bytes
beta.txt: 8 bytes
```

Files, Backups and shutil

Scripts can find files with glob("*.txt") and read sizes with stat().st_size. The shutil module handles high-level file operations:

- copyfile(): copies content only; copy(): content and permissions.
- copy2(): content, permissions and modification timestamps.
- move(): moves a file or directory; disk_usage(): total, used and free space.

Backup example: confirm the source is a file, create the backup folder, refuse to overwrite an existing backup, then call shutil.copy2().

- **Dry Run:** prints planned actions without changing any files.
- **Idempotent Operation:** gives the same result however many times it runs (e.g. mkdir(exist_ok=True)).

Workflow: Validate Source → Create Folder → Check Existing Backup → Copy

- **copyfile(src, dst):**
  Copies file content only.
- **copy(src, dst):**
  Copies content and permission flags.
- **copy2(src, dst):**
  Copies content, permissions, and preserves modification timestamps.
- **move(src, dst):**
  Recursively moves a file or directory.
- **disk_usage(path):**
  Returns total, used, and free space on the filesystem.

Quick Comparison: Which Copy Function to Use?

All three functions copy the file content. They differ in how much metadata is kept.

Function	Copies content	Copies permissions	Keeps timestamps
copyfile()	Yes	No	No
copy()	Yes	Yes	No
copy2()	Yes	Yes	Yes

Backup Tips

- Prefer copy2() for backups: the original modification time is kept, so you can tell when the data last changed.
- Check destination.exists() before copying so an older backup is never overwritten by mistake.
- Use disk_usage(".") first to confirm there is enough free space; it returns total, used and free bytes.
- Test with a dry run (print the planned copy only) before running the real backup.

```
from pathlib import Path
import shutil

source = Path("input/alpha.txt")
destination = Path("backup/alpha.txt")

if not source.is_file():
    raise FileNotFoundError("The source file is missing")

# Ensure destination directory exists
destination.parent.mkdir(parents=True, exist_ok=True)

if destination.exists():
    raise FileExistsError("The backup already exists")

shutil.copy2(source, destination)
print("Backup created:", destination)
```

System Checks and Running Commands

A disk check reads shutil.disk_usage("."), calculates the used percentage and prints a **warning at 80% or above**. To run external programs, use subprocess.run() with the command as a list so no raw shell is opened.

- Use check=True to raise an error if the command fails.
- Use capture_output=True, text=True to read the output as text.
- Use timeout and catch CalledProcessError and TimeoutExpired.

```
subprocess.run(["git", "--version"], check=True,
               capture_output=True, text=True, timeout=5)
```

```
import subprocess

try:
    result = subprocess.run(
        ["git", "---version"],
        check=True,
        capture_output=True,
        text=True,
        timeout=5,
    )
    print("Git version:", result.stdout.strip())

except subprocess.CalledProcessError as error:
    print("Command failed with return code:", error.returncode)

except subprocess.TimeoutExpired:
    print("Command timed out")
```

Logging

A report presents the result of a task (e.g. Files: 2 | Total bytes: 13), while a **log** records runtime steps, timing and errors over time. Logging levels show how serious a message is:

Level	Value	Purpose
DEBUG	10	Detailed diagnostic information for troubleshooting.
INFO	20	Normal progress and milestone messages.
WARNING	30	Unexpected event that does not stop execution.
ERROR	40	A task failed due to an exception.
CRITICAL	50	Severe failure threatening the application.

To stop log files growing endlessly, use RotatingFileHandler (rotates at a size limit) or TimedRotatingFileHandler (rotates at intervals, e.g. midnight).

```
import logging
logging.basicConfig(level=logging.INFO)
logging.info("Automation script initiated")

14. Worked Example: File Inventory with Logging
Save this script as inventory.py:

import json
import logging
from pathlib import Path
import sys

base = Path(__file__).resolve().parent

logging.basicConfig(
    filename=base / "automation.log",
    encoding="utf-8",
    level=logging.INFO,
    format="%(asctime)s %(levelname)s %(message)s",
)

def main():
    source = base / "input"
    try:
        if not source.is_dir():
            raise FileNotFoundError("The input folder is missing")
        logging.info("Inventory started")
        files = []
        for path in sorted(source.iterdir()):
            if (
                path.suffix == ".txt"
                and path.is_file()
                and not path.is_symlink()
            ):
                files.append({
                    "name": path.name,
                    "bytes": path.stat().st_size,
                })
        report = {
            "file_count": len(files),
            "files": files,
            "total_bytes": sum(item["bytes"] for item in files),
        }
        with open(base / "inventory.json", "w", encoding="utf-8") as output:
            json.dump(report, output, indent=2)
        logging.info("Inventory finished: %s files processed", len(files))
        print(f"Files: {len(files)}; bytes: {report['total_bytes']}")
        return 0
    except OSError:
        logging.exception("Inventory failed due to an OS error")
        print("Inventory failed; inspect automation.log", file=sys.stderr)
        return 1

if __name__ == "__main__":
    sys.exit(main())
```

Worked Example: File Inventory with Logging

inventory.py scans the input folder for .txt files, writes a JSON report (file_count, files, total_bytes) and records its activity in automation.log.

- Path(__file__).resolve().parent keeps paths correct from any working directory.
- is_symlink() filtering skips broken shortcuts.
- logging.exception() adds the full error traceback to the log.
- The script returns 0 on success and 1 on failure for shell runners.

Workflow: Check Folder → Scan .txt Files → Write inventory.json → Write automation.log

```python
import json
import logging
from pathlib import Path
import sys

base = Path(__file__).resolve().parent

logging.basicConfig(
    filename=base / "automation.log",
    encoding="utf-8",
    level=logging.INFO,
    format="%(asctime)s %(levelname)s %(message)s",
)


def main():
    source = base / "input"

    try:
        if not source.is_dir():
            raise FileNotFoundError("The input folder is missing")

        logging.info("Inventory started")
        files = []

        for path in sorted(source.iterdir()):
            if (
                path.suffix == ".txt"
                and path.is_file()
                and not path.is_symlink()
            ):
                files.append({
                    "name": path.name,
                    "bytes": path.stat().st_size,
                })

        report = {
            "file_count": len(files),
            "files": files,
            "total_bytes": sum(item["bytes"] for item in files),
        }

        with open(base / "inventory.json", "w", encoding="utf-8") as output:
            json.dump(report, output, indent=2)

        logging.info("Inventory finished: %s files processed", len(files))
        print(f"Files: {len(files)}; bytes: {report['total_bytes']}")
        return 0

    except OSError:
        logging.exception("Inventory failed due to an OS error")
        print("Inventory failed; inspect automation.log", file=sys.stderr)
        return 1


if __name__ == "__main__":
    sys.exit(main())
```

Sample Output of inventory.py

Running the script with two text files in the input folder produces the two files below.

inventory.json (report)

```
{
  "file_count": 2,
  "files": [
    {"name": "alpha.txt", "bytes": 5},
    {"name": "beta.txt", "bytes": 8}
  ],
  "total_bytes": 13
}
```

automation.log (log)

```
2026-10-25 10:30:01,123 INFO Inventory started
2026-10-25 10:30:01,125 INFO Inventory finished: 2 files processed
```

Key Points in the Code

- The filter checks three things: the .txt suffix, is_file() and not is_symlink().
- sorted(source.iterdir()) keeps the file order the same on every run.
- Only OSError is caught, and logging.exception() saves the full traceback.
- After running, type echo $? in the shell: 0 means success, 1 means failure.

Scheduling with Cron

Cron is a daemon that runs scheduled commands on Unix-like systems. Each crontab line has five time fields followed by the command:

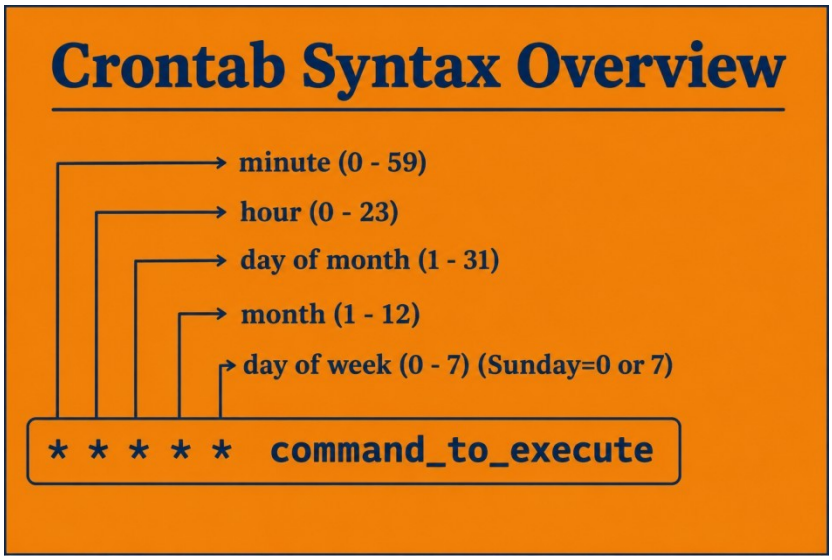
```
* * * * *  command_to_execute
| | | | |
| | | | +-- day of week (0-7, Sunday = 0 or 7)
| | | +---- month (1-12)
| | +------ day of month (1-31)
| +-------- hour (0-23)
+---------- minute (0-59)
```

- */15 * * * * : every 15 minutes.
- 30 8 * * 1-5 : 08:30 AM, Monday to Friday.
- 0 9,18 * * * : twice daily at 09:00 AM and 06:00 PM.

Running Scripts with Cron

Edit the user crontab with crontab -e. **Key rule**: call the virtual environment's interpreter by absolute path (.venv/bin/python) and redirect >> cron-output.log 2>&1 to capture errors. Common failures:

1. **Environment mismatch:** cron does not load .bashrc or shell profiles.
2. **Relative paths:** the default working directory breaks relative path logic.
3. **Missing binaries:** PATH is minimal, so use full paths.
4. **Overlapping runs:** a long task may still run when the next one starts.

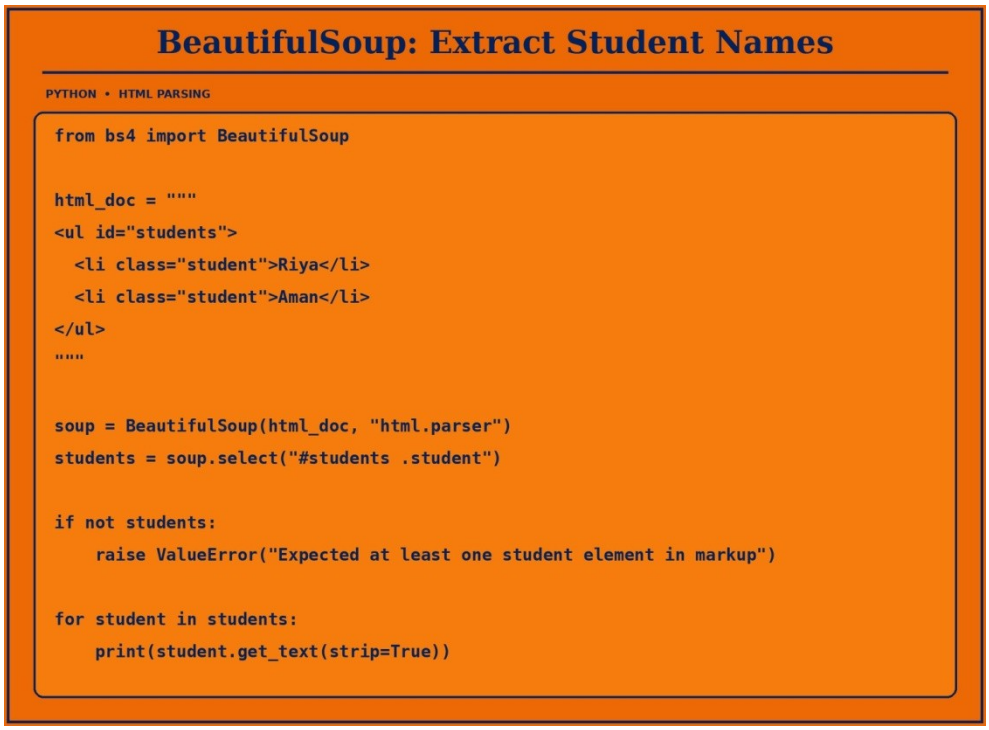


Web Scraping with Beautiful Soup

Web scraping extracts information from HTML. **Beautiful Soup** (bs4, installed with pip install beautifulsoup4) parses static HTML; it does **not** run JavaScript. Use soup.select("#students .student") to pick elements and get_text(strip=True) to read them, and raise an error if no element is found.

- Check robots.txt and respect rate limits.
- Set connection and read timeouts on requests.
- For JavaScript-driven pages, use browser automation.

Workflow: Fetch Page → Check Status (200) → Parse HTML → Validate Selectors → Export Data



Common Mistakes and Quick Revision

Common mistakes to avoid:

1. Relying on the working directory instead of Path(__file__).
2. Opening files in "w" mode before validating input.
3. Hiding errors with except: pass.
4. Using relative interpreter paths in cron jobs.
5. Letting log files grow without rotation.